English Medium

EDU TERIA

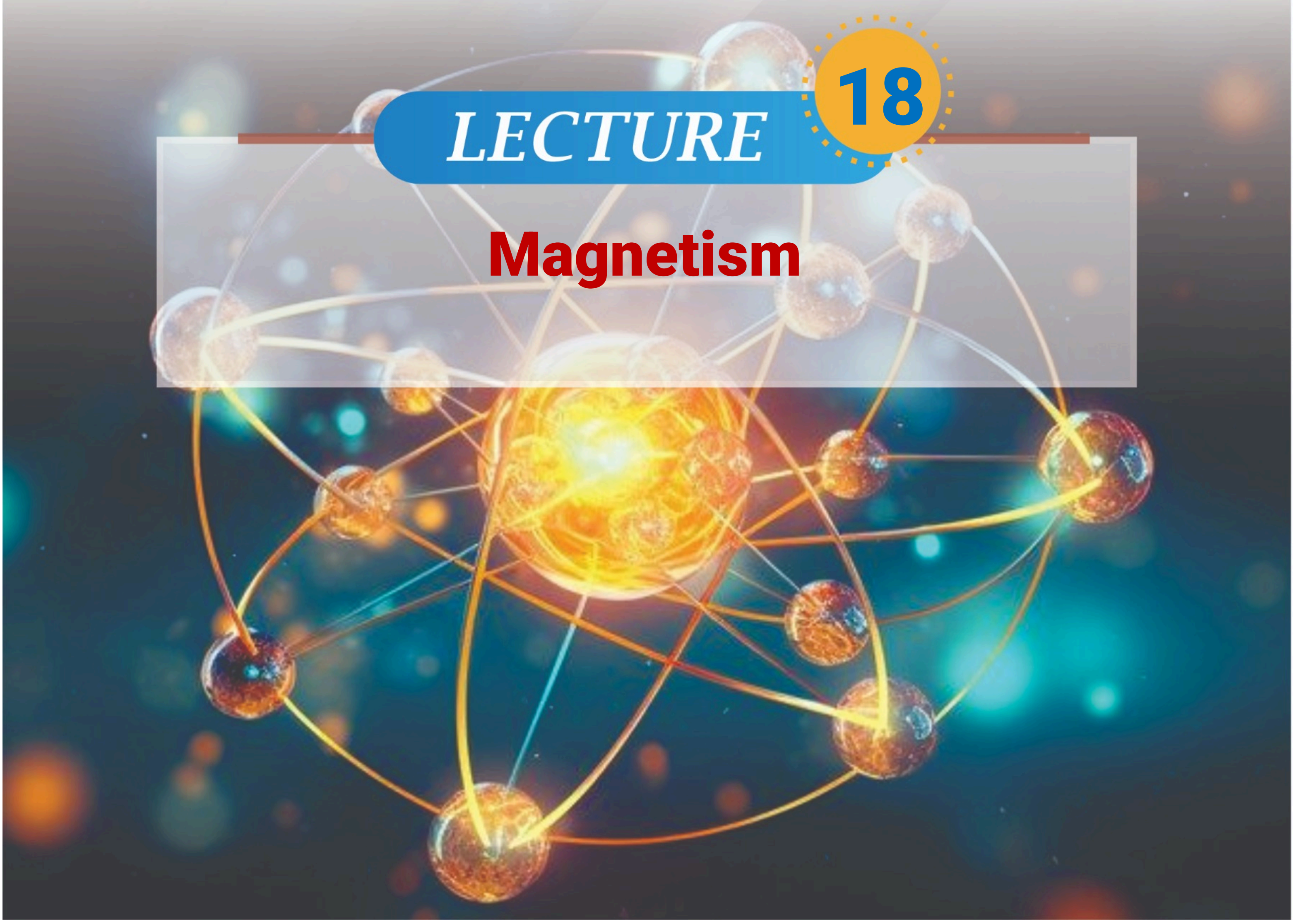
72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE 18

Magnetism



Magnetism

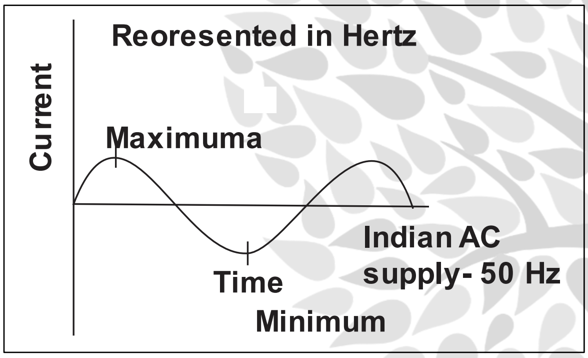
Magnetism is a physical phenomenon produced by the motion of electric charges, resulting in attractive and repulsive forces between objects.

Types of current:

Electric current is primarily classified into two types based on the direction of charge flow, Direct Current (DC), which flows in a single constant direction, and Alternating Current (AC), which reverses direction periodically

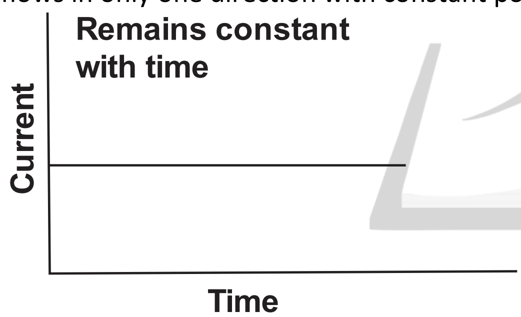
Alternating Current (AC)

Alternating current (AC) is the current that changes its direction and magnitude periodically.



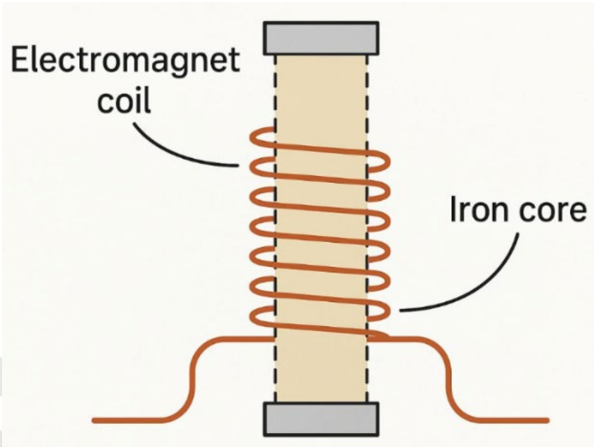
Direct Current (DC)

Direct current (DC) is the type of electric current that flows in only one direction with constant polarity.



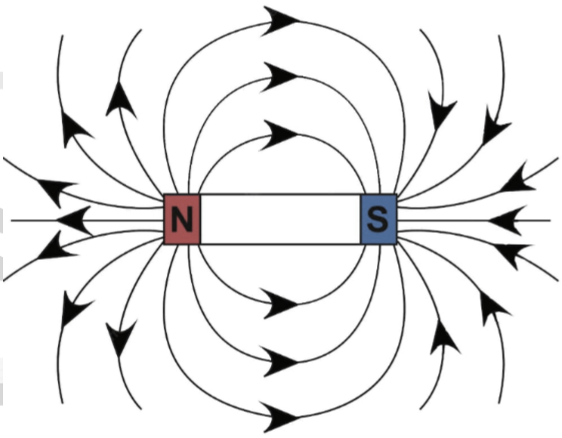
Electromagnetism

- The branch of physics which deals with the magnetism due to electric current or moving charge (i.e. electric current is equivalent to the charges or electrons in motion) is called electromagnetism.



Magnetic Field

- A region of space near a magnet, electric current or moving charged particle in which magnetic effects are exerted on any other magnet, electric current, or moving charged particle.
- It is also known as magnetic flux density or magnetic induction or magnetic field.



- **SI Unit :** weber/(meter)2 or tesla (T) C.G.S.: gauss (1 Tesla = 104 gauss)

Oersted Discovery

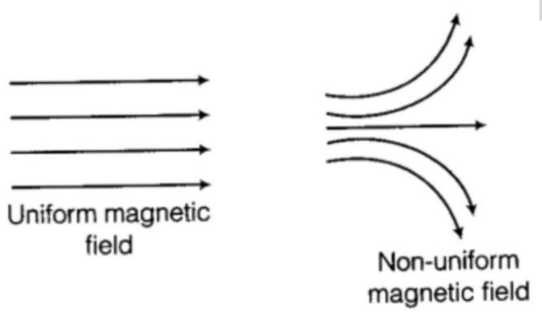
- The relation between electricity and magnetism was discovered by Oersted in 1820.

- Oersted showed that the electric current through the conducting wire deflects the magnetic needle held near the wire.
- When the direction of current conductor is reversed then deflection of magnetic needle is also reversed.
- On increasing the current in conductor or bringing the needle closer to the conductor, the deflection of magnetic needle increases.
- Oersted discovered a magnetic field around a conductor carrying electric current. Other related facts are as follows:
    - A magnet at rest produces a magnetic field around it while an Oersted's experiment.
    - Currently in the wire deflects the compass needle electric charge at rest produce an electric field around it.

Oersted discovered a magnetic field around a conductor carrying electric current. Other related facts are as follows:—

- A current carrying conductor has a magnetic field and cotem electric field around it.
- On the other hand, charge movement with a uniform velocity has an electric as well as a magnetic field around it.
- All oscillating or an accelerated charge produces E.M. waves also in addition to electric and magnetic fields.

Magnetic field lines: These are imaginary lines that represent the direction and strength of the magnetic field around a magnet. They emerge from the north pole, enter the south pole, and form closed loops inside the magnet.

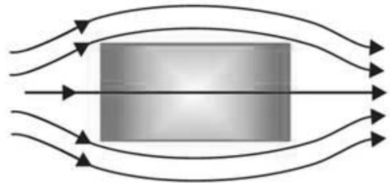


Earth as a Magnet / Earth's Magnetism: The Earth behaves like a giant magnet, and its magnetic field is spread all around it.

Curie Temperature: The temperature at which a ferromagnetic substance changes into a paramagnetic substance.

Materials ae mainly classified into three types based on their magnetic properties:

- The main types of magnetic materials are diamagnetic, paramagnetic, and ferromagnetic, which are classified based on their interaction with a magnetic field.
- Diamagnetic materials are weakly repelled, paramagnetic materials are weakly attracted, and ferromagnetic materials are strongly attracted and can retain magnetism.



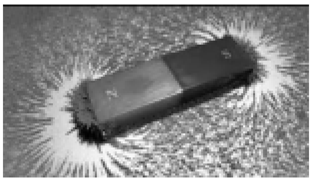
Magnetic Field lines through Diamagnetic material

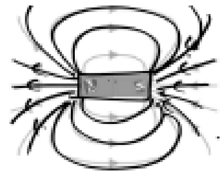
Magnetic Field lines through paramagnetic material

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Effect of magnetic field	Weakly magnetized in the direction opposite to the magnetic field	Weakly magnetized in the direction of the magnetic field	Strongly magnetized in the direction of the magnetic field
Behavior toward magnet	Weakly repelled by a magnet	Weakly attracted by a magnet	Strongly attracted by a magnet
Magnetic susceptibility (χ)	Small and negative ($\chi < 0$)	Small and positive ($\chi > 0$)	Very large and positive ($\chi >> 0$)
Examples	Bismuth, copper, gold, silver, water, salt, zinc	Manganess, aluminium, platinum, oxygen(O_2)	Iron, nickel, cobalt, steel, magnetite

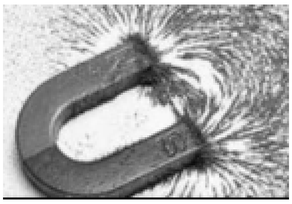
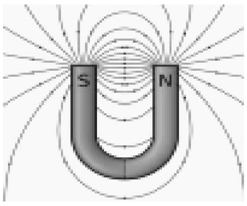
Types of Magnets:

- **Bar Magnet:** A common example used in laboratories; a bar magnet is typically ferromagnetic in nature.

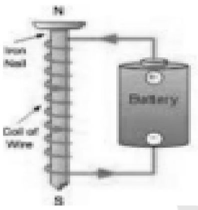


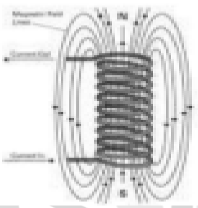


- **Horseshoe Magnet:** A U-shaped magnet designed to produce a strong magnetic field at its poles. It is another example of a permanent magnet, typically made from a ferromagnetic material.

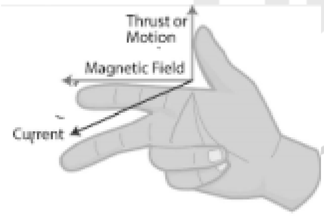



- **Electromagnet:** A coil of wire wounds around a ferromagnetic core. Its magnetism is temporary and exists only when an electric current flows through the wire.



- **Fleming's rules (Left-Hand Rule for motors, Right-Hand Rule for generators)** determine the directions of force, magnetic field, and current in electromagnetic systems. By holding the thumb, forefinger, and middle finger perpendicular to each other, you can identify how electrical energy converts to motion or vice-versa, using the FBI (Force, B-field, Induction) mnemonic



Fleming left hand thumb Rule

Direction of magnetic force on conductor.

Thumb— Direction of magnetic force

Fore Finger—Direction of Magnetic Field

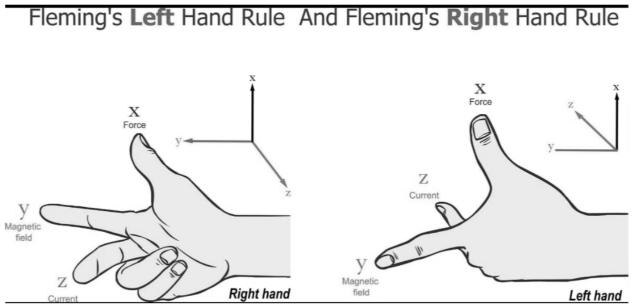
Middle finger—Direction of induced current.

Fleming right hand thumb Pule

Provides direction of motion of conductor.

Thumb— Direction of motion of conductor

Fore finger— Direction of magnetic field Middle finger Direction of induced current



Magnetic force on charge particle present in Magnetic Field.

$F = q\,v\,B\sin\theta$ $\quad \vec{F} = q(\vec{v}\times\vec{B})$

q — charge

B — Magnetic field

V — Velocity

Magnetic force upon current carrying wire present in magnetic field.

$F = B\,I\,\ell\,\sin\theta$

B = Magnetic Field

I = Induced current

ℓ = length of current carrying wire

θ = Angle between current direction and magnetic field

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